

Haematological and serum biochemical responses of rabbit does to crude *Moringa oleifera* leaf extract at gestation and lactation

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Received: 29 June 2014 / Accepted: 8 January 2015 / Published online: 17 February 2015
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Abstract As the plant *Moringa oleifera* is used in herbal medicines for animals, an experiment was carried out to assess the effects of crude *M. oleifera* leaf extract (CMOLE) on the blood profile of rabbit does during gestation and lactation. Twenty-four mature does (mean weight 2200 g) housed individually were assigned to four treatments in a completely randomised design. The animals in treatments 2, 3 and 4 were orally given 100, 200 and 300 mL/L CMOLE, respectively, at 2.5 mL/kg body weight at 48 h intervals for 9 weeks. The control animals (treatment 1) were given with water only. All the does were mated with untreated bucks 2 weeks into the experiment. Blood samples were collected at 3rd trimester (day 25 of gestation) and 2nd week of lactation. During gestation, levels of erythrocytes, leukocytes, haematocrit, haemoglobin, lymphocytes, monocytes and eosinophils were not significantly different among the treatments. However, animals on treatment 2 had the highest platelets ($148.8 \times 10^9/L$), not significantly different from those on treatments 3 ($141.5 \times 10^9/L$) and 4 ($135.0 \times 10^9/L$), but higher ($p < 0.05$) than the control ($126.6 \times 10^9/L$). Haematological parameters during lactation were not significantly different among the treatments. Of the serum biochemical variables examined during gestation, only urea was higher ($p < 0.05$) in control rabbits than those administered with 300 mL/L CMOLE. In lactation, only cholesterol was significantly ($p < 0.05$) reduced with an increase in CMOLE concentration. This suggests that *Moringa* has a hypocholesterolemic effect and is safe for use up to 300 mL/L for both nutritional and medicinal purposes.

Keywords Rabbit does · *Moringa* extract · Blood profile · Hypocholesterolemia

Introduction

Moringa oleifera provides a rich and rare combination of zeatin, quercetin, kaempferol and many other photochemical (Farooq et al. 2007). It is very important for its medicinal value in animals (Takano 2004). Various parts of the plant such as the leaves, roots, seed, bark, fruit, flowers and immature pods act as cardiac and circulatory stimulants, possess antitumour (Rockwood et al. 2013), antipyretic, antiepileptic, anti-inflammatory, and antiulcer properties (Fahey 2005). Other important medicinal properties of the plant include antispasmodic, diuretic, antihypertensive, cholesterol lowering, antioxidant, antidiabetic, hepatoprotective, antibacterial and antifungal activities (Amaglo et al. 2010; Gowrishankar et al. 2010; Mbikay 2012). *M. oleifera* parts are being employed for the treatment of different ailments in the indigenous system of medicine (Rockwood et al. 2013), particularly in South Asia (Farooq et al. 2007).

Extracts of various *Moringa* tissues have been used as antimicrobial, anti-inflammatory and hepatoprotective (Fahey 2005), anticancer (Berkovich et al. 2013) and antitrypanosomal (Jabeen et al. 2008) agents. In addition, leaf extracts have been shown to regulate thyroid status (Tahiliani and Kar 2000) and cholesterol levels in rats (Ghasi et al. 2000). Recently, novel derivatives of thiocarbamates and nitriles which stimulate insulin release in animals have been found in the plant, explaining its antidiabetic mechanism of action (Fahey 2005; Mbikay 2012). Indeed, science continues

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to validate the ancient traditional therapeutic uses of *Moringa*. Ahemen et al. (2013) observed alteration in blood profile of growing rabbits fed with graded levels of *Moringa* leaf meal as a replacement to soya bean meal. Blood examination gives the opportunity to investigate the presence of several metabolites and other constituents and helps detect conditions of stress, which can be nutritional, environmental or physical (Aderemi 2004). In view of the nutritional and medicinal benefits of *M. oleifera* in animals, this study was designed to assess the effect of administration of crude leaf extract of *M. oleifera* when administered as a supplement on haematological and serum biochemical response of rabbit does at gestation and lactation.

Materials and methods

Experimental location

The trial which lasted 9 weeks was conducted at the Rabbitry Unit of the Teaching and Research Farm, University of Ibadan, Ibadan, Nigeria (Latitude 7° 20' N and 3° 50' E, 200 m above the sea level).

Sample extract preparation

The *Moringa* plant (*M. oleifera*) was properly identified by an agronomist in the Department of Agronomy, University of Ibadan, and was collected from the garden of the same department. Fresh leaves were removed from the branches, washed and pounded with mortar and pestle. The pounded leaves were then squeezed and sieved in order to remove the crude extract into a clean container before diluting with distilled water of known volume. This was done so as to obtain four treatments containing 0, 100, 200 and 300 mL of CMOLE per litre of distilled water representing treatments 1 (control), 2, 3 and 4, respectively. This procedure was carried out every 2 days until the end of the experiment.

Experimental animals and management

Twenty-four mature crossbreed (New Zealand White and Chinchilla) rabbit does weighing between 2000 to 2550 g were procured from a rabbit farm within Ibadan metropolis in Oyo State. They were housed individually in experimental cages and were randomly divided into four treatment groups on weight basis, with six replicates per treatment in a completely randomised design. The animals were allowed to acclimatise for a period of 2 weeks.

The animals were offered the same feed containing 16 % crude protein, 10 % crude fibre and 2700 kcal/kg digestible energy ad libitum. Fresh water was made available always for the animals. The administration of the extract to the animals

was carried out every 48 h for 9 weeks at a dose of 2.5 mL/kg body weight through oral drench. All the does were mated with untreated bucks 2 weeks into the experiment and allowed to gestate and lactate after kindling.

Blood collection and evaluation

First blood collection was done at 3rd trimester, while the 2nd blood collection was carried out during 2nd week of lactation. The blood was sampled through the jugular vein of each rabbits into two sets of sample bottles each, one bottle containing appropriate quantity of ethylene diamine tetra-acetic acid for haematological analysis, and the other bottle without anticoagulant for serum analysis. Packed cell volume, haemoglobin, leukocyte and erythrocyte counts were determined as outlined in Ewuola and Egbunike (2008). The differential leukocyte count was done as described by Schalm (1975). The mean corpuscular volume (MCV), mean corpuscular haemoglobin (MCH) and mean corpuscular haemoglobin concentration (MCHC) were calculated using the standard formulae as modified by Thrall and Weiser (2002). Total protein was determined as described by Peter et al. (1982). Serum albumin was analysed according to the method described by Dumas et al. (1971) while globulin was calculated using the formula: total protein – albumin. Serum urea and creatinine were analysed by the method described by Foster and Hochhlozer (1971) and Blass et al. (1974), respectively. Serum cholesterol and ALP were determined by the methods of Flegg (1973) and Klein et al. (1960), respectively. The AST and ALT were determined using the method described by Cain (1976).

Data analysis

All data were tested using a one-way analysis of variance (SAS 2003), and Duncan's multiple range tests was used to separate means.

Results

Haematological response during gestation and lactation

The haematological parameters of rabbit does administered CMOLE at gestation period are shown in Table 1. Haematocrit, haemoglobin (Hb), erythrocyte (RBC), leukocyte (WBC), MCH, MCHC, lymphocytes, monocytes and eosinophil values were not significantly different among the treatments. However, MCV, platelets and neutrophils were significantly ($p < 0.05$) influenced by the treatments. The MCV of rabbits on T2, T3 and T4 was significantly ($p < 0.05$) higher than those on T1 (control). The platelet of rabbits on T2 had the highest mean value ($148.8 \times 10^9/L$)

Table 1 Haematological parameters of mature rabbit does administered with crude *Moringa oleifera* leaf extract at gestation period

Parameters	T1	T2	T3	T4	SEM
Erythrocyte ($\times 10^{12}/L$)	4.21	3.75	3.97	3.93	0.17
Leukocyte ($\times 10^9/L$)	5.8	5.3	5.6	6.1	0.41
Haematocrit (L)	0.37	0.36	0.37	0.38	0.01
Haemoglobin (mmol/L)	7.7	7.5	7.7	7.8	0.22
MCV (fL)	82.2 ^b	93.4 ^a	91.7 ^a	91.5 ^a	2.30
MCH (fmol)	1.77	1.87	1.86	1.83	0.06
MCHC (mmol/L)	20.68	20.68	20.67	20.67	0.002
Platelets ($\times 10^9/L$)	127 ^b	149 ^a	142 ^{a, b}	135 ^{a, b}	10.8
Lymphocytes ($\times 10^9/L$)	3.2	3.9	3.2	2.7	0.5
Neutrophils ($\times 10^9/L$)	2.4 ^{a, b}	1.2 ^b	2.1 ^{a, b}	3.1 ^a	0.4
Monocytes ($\times 10^8/L$)	1.0	1.4	1.6	1.1	0.2
Eosinophils ($\times 10^8/L$)	1.3	1.3	1.6	1.8	0.3

MCV mean corpuscular volume, MCH mean corpuscular haemoglobin, MCHC mean corpuscular haemoglobin concentration, T1 control (water only), T2 100 mL CMOLE/L, T3 200 mL CMOLE/L, T4 300 mL CMOLE/L, CMOLE crude *Moringa oleifera* leaf extract

^{a, b} Means along the same row with different superscripts are significantly ($p < 0.05$) different

which was significantly ($p < 0.05$) higher than those on the control ($126.6 \times 10^9/L$). The neutrophils of animals on T4 had the highest mean value ($3.14 \times 10^9/L$) and was significantly ($p < 0.05$) higher than those on T2. The haematological parameters of does administered with CMOLE during lactation are shown in Table 2. There were no significant differences among the treatments for all the parameters (erythrocyte, leukocyte, haematocrit, haemoglobin, MCV, MCH, MCHC, platelet, lymphocyte, neutrophil, monocyte and eosinophil) measured.

Table 2 Haematological parameters of mature rabbit does administered with crude *Moringa oleifera* leaf extract at lactation period

Parameters	T1	T2	T3	T4	SEM
Erythrocyte ($\times 10^{12}/L$)	3.59	3.38	3.60	3.56	0.24
Leukocyte ($\times 10^9/L$)	6.1	6.9	8.0	7.8	1.2
Haematocrit (L)	0.38	0.33	0.36	0.31	0.02
Haemoglobin (mmol/L)	7.8	6.8	7.4	6.5	0.4
MCV (fL)	85.1	91.5	89.5	85.3	2.6
MCH (fmol)	1.94	1.89	1.85	1.70	0.08
MCHC (mmol/L)	20.68	20.68	20.68	20.69	0.002
Platelets ($\times 10^9/L$)	93	108	125	109	15.5
Lymphocytes ($\times 10^9/L$)	2.7	3.6	4.3	3.5	0.8
Neutrophils ($\times 10^9/L$)	2.9	2.5	3.2	3.4	0.7
Monocytes ($\times 10^8/L$)	1.7	1.2	1.8	2.1	0.5
Eosinophils ($\times 10^8/L$)	3.6	1.3	1.6	1.8	2.3

MCV mean corpuscular volume, MCH mean corpuscular haemoglobin, MCHC mean corpuscular haemoglobin concentration, T1 control (water only), T2 100 mL CMOLE/L, T3 200 mL CMOLE/L, T4 300 mL CMOLE/L, CMOLE crude *Moringa oleifera* leaf extract

Serum biochemical response during gestation and lactation

The serum biochemical response of the rabbit does to CMOLE during pregnancy is shown in Table 3. The total protein, albumin, globulin, albumin/globulin ratio, cholesterol, creatinine, alanine amino transferase, aspartate amino transferase and alkaline phosphatase values were not significantly different among the treatments. However, serum urea level in rabbits on T1 had the highest mean value (19.04 mmol/L) similar with those on T2 (16.36 mmol/L)

Table 3 Serum biochemical response of rabbit does to crude *Moringa oleifera* leaf extract during pregnancy

Parameters	T1	T2	T3	T4	SEM
Total protein (mmol/L)	0.072	0.072	0.071	0.068	0.00016
Albumin (mmol/L)	0.02	0.01	0.02	0.03	0.00074
Globulin (mmol/L)	0.059	0.058	0.056	0.065	0.00035
Albumin/globulin	0.3	0.2	0.3	0.2	0.07
Cholesterol (mmol/L)	1.6	1.4	1.6	1.8	0.3
Urea (mmol/L)	19.0 ^a	16.4 ^{a, b}	16.1 ^{a, b}	15.2 ^b	0.8
Creatinine ($\mu\text{mol/L}$)	31.8	31.8	36.2	37.3	3.9
ALT (μmol)	0.20	0.22	0.19	0.16	0.07
AST (μmol)	0.08	0.11	0.11	0.11	0.01
ALP (nmols ⁻¹ /L)	3.3	3.2	2.9	2.8	0.3

ALT alanine amino transferase, AST aspartate amino transferase, ALP alkaline phosphatase, SEM standard error of mean, T1 control (water only), T2 100 mL CMOLE/L, T3 200 mL CMOLE/L, T4 300 mL CMOLE/L, CMOLE crude *Moringa oleifera* leaf extract

^{a, b} Means along the same row with different superscripts are significantly ($p < 0.05$) different

and T3 (16.06 mmol/L), but significantly ($p<0.05$) higher than that of rabbits on T4 (15.16 mmol/L).

The serum biochemical response of the rabbit does to CMOLE during lactation is shown in Table 4. The total protein, albumin, globulin, albumin/globulin ratio, urea, creatinine, ALT, AST and ALP were not significantly different among the treatments. However, the control rabbits that received no extract had the similar cholesterol level with those on T2 and T3, but significantly ($p<0.05$) higher than those on T4 which had the least mean value.

Discussion

Haematological response during gestation and lactation

The result of the haematology at gestation period showed that the mean values of most haematological parameters examined were not significantly different among the treatments. This finding was at variance with the work of Adedapo et al. (2009) who reported that oral treatment of rats with aqueous *Moringa oleifera* leaf extract at 400, 800 and 1600 mg/kg caused varied significant changes in the total RBC, PCV, Hb, MCHC and total and differential WBC probably because of the higher concentration of the extract used in their study. Apparently, the values of haematocrit, haemoglobin, leukocytes, neutrophils, lymphocytes and MCHC obtained from this study were within the normal range as reported by Mitruka and Rawnsley (1977) which was an indication that the leaf extract had no detrimental effect on the animals as also observed by Ahemen et al. (2013). The increase in MCV

values could be an indication that the leaf extract was able to increase the size of the erythrocyte, thus aiding and improving the erythrocyte morphologically.

Nonsignificant differences that exist in the haematological parameters of lactating rabbits examined revealed that there was no negative effect on the health status of the animals. This finding agrees with the work of Nikkon et al. (2009) and Ahemen et al. (2013) who reported no significant differences in erythrocyte, leukocyte, haemoglobin, leukocyte differential counts and platelets of rats administered with *Moringa* bark extract. However, Adedapo et al. (2009) reported that oral treatments of rats with aqueous *M. oleifera* leaf extract at 400, 800 and 1600 mg/kg caused varied significant changes in the values of these parameters. The leukocyte and leukocyte differential count were within the normal physiological range recommended for healthy rabbits except for eosinophils and monocytes. The rabbits could be adjudged to have good immune status.

Serum biochemical response of gestating and lactating does

The serum biochemical response of gestating rabbit does to crude MOLE revealed that the total protein for all the treatments were within the normal range (0.06–0.083 mmol/L) reported by Mitruka and Rawnsley (1977), while albumin and globulin were also similar among the treatments indicating no adverse effect on the animals. This agreed with Ghasi et al. (2000) and Ahemen et al. (2013) who reported no significant difference in serum proteins of rats and rabbits administered crude *Moringa* leaf extract and meal, respectively. Urea is a waste product of protein metabolism, and higher value of serum urea may be an indication of inefficient utilisation of protein supplied in the diet. The level was, however, observed to decrease significantly as the concentration of the extract administered to the animals increase, and this could be that the extract improved the utilisation of dietary protein. This corroborates the observation of Adedapo et al. (2009) who reported that graded level of aqueous extract of *Moringa* leaves causes significant decrease in urea in rats.

The nonsignificant creatinine, ALT, AST and ALP levels in the serum, during gestation, were an indication of no obvious damage to muscle and organs such as the liver and kidney probably because the level used was still below 1585 mg/kg considered as LD50 for oral administration of aqueous leaf extract of *Moringa* (Awodele et al. 2012). This agrees with the findings of Nikkon et al. (2009) who reported no significant differences in ALT, AST and creatinine levels in the serum of rats administered with *Moringa* root extract. However, Adedapo et al. (2009) reported that 400 and 600 mg/kg doses of *Moringa* leaf extract caused a significant increase in the levels of ALT and AST as also evidence in Nikkon et al. (2009) and Burali et al. (2010).

Table 4 Serum biochemical response of rabbit does to crude *Moringa oleifera* leaf extract during lactation

Parameters	T1	T2	T3	T4	SEM
Total protein (mmol/L)	0.068	0.068	0.068	0.068	0.001
Albumin (mmol/L)	0.038	0.037	0.024	0.030	0.0073
Globulin (mmol/L)	0.04	0.04	0.06	0.05	0.01
Albumin/globulin ratio	1.0	0.8	0.5	0.8	0.2
Cholesterol (mmol/L)	4.1 ^a	3.7 ^{a, b}	2.2 ^{a, b}	1.6 ^b	0.5
Urea (mmol/L)	18.0	16.1	15.5	15.2	0.9
Creatinine (μmol/L)	42	72	66	60	12.6
ALT (μmol)	0.3	0.3	0.3	0.2	0.1
AST (μmol)	0.36	0.36	0.40	0.48	0.1
ALP (nmols ⁻¹ /L)	1.0	1.3	1.2	2.2	0.6

ALT alanine amino transferase, AST aspartate amino transferase, ALP alkaline phosphatase, SEM standard error of mean, T1 control (water only), T2 100 mL CMOLE/L, T3 200 mL CMOLE/L, T4 300 mL CMOLE/L, CMOLE crude *Moringa oleifera* leaf extract

^{a, b} Means along the same row with different superscripts are significantly ($p<0.05$) different

The total protein, albumin and globulin of the lactating does were within the normal range for normal rabbits (0.06–0.083 mmol/L) (Mitruka and Rawnsley 1977) in all the treatments. This is an indication that the dietary protein was effectively absorbed and utilised by the animals, which implied that the rabbits did not suffer from hypo- or hyperproteinemia. However, the serum cholesterol levels were significantly reduced as the concentration of the extract administered increases. This showed that the leaf extract was effective in reducing circulatory cholesterol (Ghasi et al. 2000). The hypocholesterolemic effect of the extract could be attributed to the presence of β -sitosterol, a bioactive phytoconstituent present in *Moringa* leaves (Chumark et al. 2008; Mbikay 2012). The urea level in the animals was not significantly different among the treatments as also observed by Nikkon et al. (2009).

The levels of ALT and AST were identical among the treatments, suggesting proper liver function without the possibility of hepatic damage. This result is in line with the work of Pari and Kumar (2002) and Fakurazi et al. (2008) who reported the hepatoprotective activity of *M. oleifera* on anti-tubercular drug-induced liver damage in rats.

Conclusion

The study showed that administration of crude *M. oleifera* leaf extract to mature rabbit does even at concentration of 300 mL/L for a period of 9 weeks has no adverse effect on their haematological and serum biochemical responses but has hypocholesterolemic effect in lactating does.

Conflict of interest The authors declare that there is no conflict of interest that could be perceived as prejudicing the impartiality of the research reported.

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